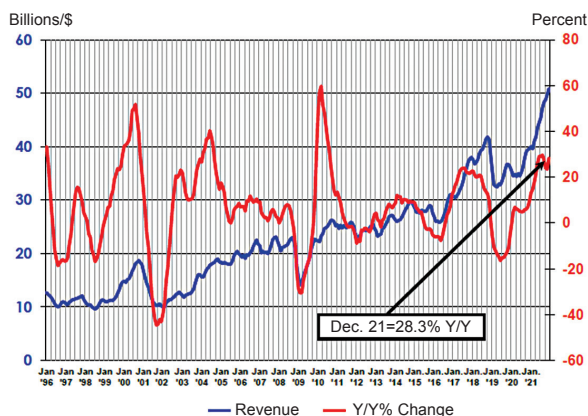


THE DASHBOARD



Global Chip Market Saw Highest-Ever Annual Sales Total in 2021

Global semiconductor sales totaled \$555.9 billion in 2021, the highest-ever annual total and an increase of 26.2% compared to the 2020 total of \$440.4 billion. China remained the largest individual market for semiconductors, with sales there totaling \$192.5 billion in 2021, an increase of 271%. Analog had the highest annual growth rate of 33.1%, reaching \$74 billion in 2021 sales. Logic (\$154.8 billion in 2021 sales) and memory (\$153.8 billion) were the largest semiconductor categories by sales.



(Source: www.semiconductors.org)

Top 10 Chip Buyers Increased Chip Spending by 25.2% in 2021

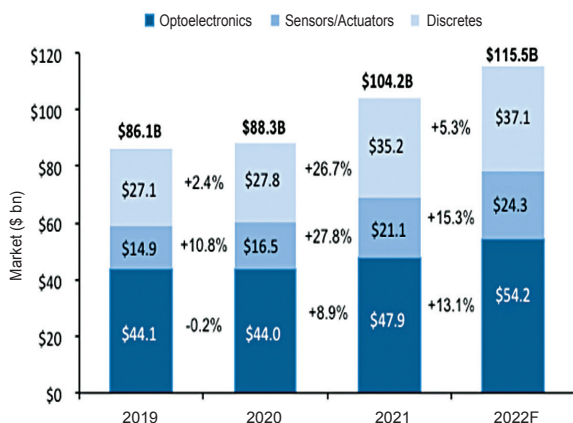
2021 Rank	2020 Rank	Vendor	2021 Spending	2021 Market Share (%)	2020 Spending	2020-2021 Growth (%)
1	1	Apple	68,269	11.7	54,180	26.0
2	2	Samsung Electronics	45,775	7.8	35,622	28.5
3	4	Lenovo	25,283	4.3	19,023	32.9
4	6	BBK Electronics	23,350	4.0	14,258	63.8
5	5	Dell Technologies	21,092	3.6	16,814	25.4
6	8	Xiaomi	17,251	3.0	10,254	68.2
7	3	Huawei	15,382	2.6	22,710	-32.3
8	7	HP Inc.	13,789	2.4	10,745	28.3
9	9	Hon Hai Precision	8,855	1.5	7,387	19.9
10	10	Hewlett-Packard Enterprise	6,736	1.2	5,395	24.8
		Others (outside top 10)	337,695	57.9	269,849	25.1
		Total semiconductor	583,477	100.0	466,237	25.1

Top 10 original equipment manufacturers (OEMs) increased their chip spending by 25.2% and accounted for 42.1% of the total market in 2021. The average selling prices (ASPs) of semiconductor chips and a wide variety of application-specific semiconductors increased by 15% or more in 2021. Apple remained at the top, having increased its spending on memory by 36.8% and on non-memory chips by 20.2% in 2021.

(Source: www.gartner.com/en)

Sensor/Actuator and Discrete Sales Surge in 2021

Total optoelectronics, sensors and actuators, and discrete semiconductors (O-S-D) revenues crossed \$100 billion for the first time ever in 2021, growing 18% to \$104.2 billion compared to \$88.3 billion in 2020, when combined sales in the three markets increased by less than 3% as the virus first took hold. Optoelectronics sales growth was muted in the 2021 rebound due to the lack of a surge in CMOS image sensors.



(Source: IC Insights)

India-Based Vendors Dominate Truly Wireless (TWS) Market in 2021

(Shipments are in thousands of units)

Company	2021 Market Share	2020 Market Share	Year-on-Year unit change (2021 vs 2020)
1. Imagine Marketing (BoAt)	39.3%	23.9%	187.9%
2. Realme	7.7%	13.1%	2.8%
3. Nexxbase (Noise)	7.5%	4.5%	194.7%
4. Palred (pTron)	6.0%	7.6%	36.8%
5. Samsung	4.7%	10.6%	-23.0%
Others	34.8%	40.3%	50.6%
Total	100%	100%	74.7%

India's TWS market witnessed a record year, with shipments reaching 20.3 million units and growing 74.7% YoY (year-on-year) in 2021. Year 2021 saw India-based vendors capture over 65% shipments in the TWS category, compared to about half of the market last year. BoAt emerged as the leader with two-fifth of the shipments during the year, which is more than the combined share of the next seven vendors.

(Source: www.idc.com)